

Ksenia Bestuzheva

<https://kbestuzheva.github.io>

Optimisation Researcher

bestuzheva@zib.de

Mathematical optimisation researcher working at the intersection of fundamental algorithmic research and computational practice in mixed-integer linear and nonlinear optimisation. Research interests include branch-and-bound methods, relaxations and cutting planes, optimality conditions, algorithm and mathematical software design, and the development of optimisation methods for challenging real-world problems as well as of general-purpose optimisation solvers.

EDUCATION

Ph.D. in computer science, *Australian National University* (Canberra, Australia) 2015-2019
Thesis title: Global Optimisation for Energy Systems
Supervisors: Dr. Hassan Hijazi, Prof. Sylvie Thiebaux, Prof. Markus Hegland

Diplom in applied mathematics and computer science, *State Management University* (Moscow, Russia) 2009-2014
With highest distinction

WORK EXPERIENCE

Full-time academic appointment, *Free University of Berlin* (Berlin, Germany) Winter semester 2026/2027

- teaching the courses "Linear Algebra II" and "Numerical Mathematics I", research, academic service

Guest lecturer, *Free University of Berlin* (Berlin, Germany) Summer semester 2026

- teaching the course "Linear Algebra I" (two lectures and one exercise session per week)

Senior software developer, *GAMS Software GmbH* (Berlin, Germany) 2025-2026

- solver development, optimisation research

Head of research group, *Zuse Institute Berlin & Research Campus MODAL* (Berlin, Germany) 2023-2025

- leading the group's research activities in mixed-integer linear and nonlinear optimisation
- head of development of the SCIP Optimization Suite, leading a developer team distributed across universities and research organisations
- head of SynLab within Research Campus MODAL, coordinating research activities, grant applications and reporting

Postdoctoral researcher, *Zuse Institute Berlin* (Berlin, Germany) 2018-2023

- research in the area of mixed-integer nonlinear programming
- development of the solver SCIP
- MSc student and intern supervision

PUBLICATIONS

Journal publications

My Google Scholar profile can be found at <https://scholar.google.com/citations?user=MwqkeJ8AAAAJ>.

- K. Bestuzheva. On local optima of box-constrained quadratic programs. (*In preparation*)
- M. Ramgraber and K. Bestuzheva. Bounding the null space: Interval-based uncertainty quantification for non-identifiable groundwater models. 2026 (<https://arxiv.org/abs/2606.10836>)
- J. Dionísio, A. Gleixner, J. P. Pedroso, and K. Bestuzheva. Branch and price for nonlinear production-maintenance scheduling in complex machinery. (*Under review in Annals of Operations Research*) (<https://arxiv.org/abs/2511.07143>)

- K. Bestuzheva, A. Gleixner, and T. Achterberg. Efficient separation of RLT cuts for implicit and explicit bilinear terms. *Mathematical Programming B*, 2024. doi: 10.1007/s10107-024-02104-0 (<https://arxiv.org/abs/2211.13545>, 8 citations together with the corresponding conference publication)
- K. Bestuzheva, A. Chmiela, B. Müller, F. Serrano, S. Vigerske, and F. Wegscheider. Global optimization of mixed-integer nonlinear programs with SCIP 8. *Journal of Global Optimization*, 2023. doi: 10.1007/s10898-023-01345-1 (<https://arxiv.org/abs/2301.00587>, 54 citations)
- K. Bestuzheva, A. Gleixner, and S. Vigerske. A computational study of perspective cuts. *Mathematical Programming Computation*, 15(4):703–731, 2023. doi: 10.1007/s12532-023-00246-4 (<https://arxiv.org/abs/2103.09573>, 14 citations)
- K. Bestuzheva, M. Besançon, W.-K. Chen, A. Chmiela, T. Donkiewicz, J. van Doornmalen, L. Eifler, O. Gaul, G. Gamrath, A. Gleixner, et al. Enabling research through the SCIP Optimization Suite 8.0. *ACM Transactions on Mathematical Software*, 49(2):1–21, 2023. doi: 10.1145/3585516 (<https://arxiv.org/abs/2303.07101>, 101 citations)
- E. Ramin, K. Bestuzheva, C. L. Gargalo, D. Ramin, C. Schneider, P. Ramin, X. Flores-Alsina, M. M. Andersen, and K. V. Gernaey. Incremental design of water symbiosis networks with prior knowledge: The case of an industrial park in Kenya. *Science of The Total Environment*, 751:141706, 2021. doi: 10.1016/j.scitotenv.2020.141706 (https://backend.orbit.dtu.dk/ws/portalfiles/portal/236329059/STOTEN_D_20_14264R2_accepted.pdf, 30 citations)
- K. Bestuzheva, H. Hijazi, and C. Coffrin. Convex relaxations for quadratic on/off constraints and applications to optimal transmission switching. *INFORMS Journal on Computing*, 32(3):682–696, 2020. doi: 10.1287/ijoc.2019.0900 (<https://optimization-online.org/wp-content/uploads/2016/07/5565.pdf>, 38 citations)
- K. Bestuzheva and H. Hijazi. Inxv optimization revisited. *Journal of Global Optimization*, 74(4):753–782, 2019. doi: 10.1007/s10898-018-0650-1 (<https://arxiv.org/abs/1707.01554>, 9 citations)

Conference publications

- L. Xu, G. Mexi, and K. Bestuzheva. Sparsity-driven aggregation of mixed integer programs. In *23rd International Symposium on Experimental Algorithms (SEA 2025)*, pages 27:1–27:15, 2025. doi: <https://doi.org/10.4230/LIPIcs.SEA.2025.27> (<https://arxiv.org/abs/2502.01192>)
- K. Bestuzheva, A. Gleixner, and T. Achterberg. Efficient separation of RLT cuts for implicit and explicit bilinear products. In *Integer Programming and Combinatorial Optimization*, pages 14–28, Cham, 2023. Springer. doi: 10.1007/978-3-031-32726-1_2
- K. Bestuzheva, A. Gleixner, and H. Völker. Strengthening SONC relaxations with constraints derived from variable bounds. In *Proceedings of Proceedings of the Hungarian Global Optimization Workshop HUGO 2022*, pages 41–44, 2022

Other publications

- G. Mexi, D. Kamp, Y. Shinano, S. Pu, A. Hoen, K. Bestuzheva, C. Hojny, M. Walter, M. E. Pfetsch, S. Pokutta, and T. Koch. State-of-the-art methods for pseudo-boolean solving with SCIP. 2025 (<https://arxiv.org/abs/2501.03390>, 3 citations)
- S. Bolusani, M. Besançon, K. Bestuzheva, A. Chmiela, J. Dionísio, T. Donkiewicz, J. van Doornmalen, L. Eifler, M. Ghannam, A. Gleixner, et al. The SCIP Optimization Suite 9.0. 2024 (<https://arxiv.org/pdf/2402.17702>, 217 citations)
- K. Bestuzheva, A. Gleixner, and H. Völker. Strengthening SONC relaxations with constraints derived from variable bounds. 2022 (<https://arxiv.org/abs/2304.12145>, 1 citation)
- K. Bestuzheva, M. Besançon, W.-K. Chen, A. Chmiela, T. Donkiewicz, J. van Doornmalen, L. Eifler, O. Gaul, G. Gamrath, A. Gleixner, et al. The SCIP Optimization Suite 8.0. 2021 (<https://arxiv.org/pdf/2112.08872>, 388 citations)
- G. Gamrath, D. Anderson, K. Bestuzheva, W.-K. Chen, L. Eifler, M. Gasse, P. Gemander, A. Gleixner, L. Gottwald, K. Halbig, et al. The SCIP Optimization Suite 7.0. 2020 (<https://www.or.rwth-aachen.de/files/research/repORt/scip70.pdf>, 331 citations)

TEACHING EXPERIENCE

Theses completed under my supervision

- F. Geis. Selecting branching rules for MINLPs via machine learning. Master's thesis, Technical University of Berlin, 2025
Supervised jointly with Dr. Timo Berthold.
This work considers nonconvex mixed-integer nonlinear programs, where branch-and-bound algorithms have to branch on both integer and continuous variables in order to find a global optimum. The thesis proposes a machine learning approach for deciding, for a given instance, whether branching is to be performed on integer variables first, or whether continuous variables should be considered for branching together with integer variables.
- H. Müller. Algorithms for constrained optimization using sums of nonnegative circuit polynomials certificates. Master's thesis, Humboldt University of Berlin, 2021
Supervised jointly with Prof. Andrea Walther.
The thesis introduced a new branch-and-bound framework for global polynomial optimisation that employs Sums of Nonnegative Circuit polynomials (SONC) nonnegativity certificates to obtain dual bounds at nodes of the branch-and-bound tree.

Teaching assignments

- Course "Linear Algebra I" at the Free University of Berlin in the summer semester of 2026
- Seminar on discrete optimisation and machine learning at the Technical University of Berlin
(<https://sites.google.com/view/seminardoml-rose2021/previous-semesters/rose-2024?authuser=0>)

Other teaching experience

- Lectures and interactive sessions on mixed-integer nonlinear programming at the summer schools Computational Optimization at Work 2020 and 2024 (lecture recordings available online at <https://www.youtube.com/watch?v=BRe8855C4BA> and <https://www.youtube.com/watch?v=m3LQep1qYA4>, respectively)
- Co-supervising an ongoing Ph.D. thesis on column generation approaches to production-maintenance scheduling
- Supervision of international undergraduate interns
- Participation as a mentor in Google Summer of Code projects

INVITED RESEARCH WORKSHOPS

- Combinatorial Optimization Workshop (Aussois, 2023) - invited participant
- Mixed Integer Programming Workshop (Los Angeles, 2023) - invited speaker
- Workshop "Discrete Nonlinear Optimization: Mathematics and Algorithms" (Oberwolfach, 2027) - invited participant

AWARDS

- COIN-OR Cup: Hassan Hijazi, Guanglei Wang, Ksenia Bestuzheva, Smitha Gopinath, Mertcan Yetkin, and Carleton Coffrin (Gravity) (03.11.2021)
- Best Student Paper Award, 24th National Conference of the Australian Society for Operations Research (18.11.2016)

OTHER RELATED EXPERIENCE

Scientific committee member , <i>Global Optimization Workshop</i>	2025
Technical editor , <i>Mathematical Programming Computation</i>	2022-present
Performing technical reviews of submitted articles focusing on the software and/or data aspect of the submission	
Program committee member , <i>Symposium on Experimental Algorithms</i>	2024
Contributor , <i>ASCEND - mathematical modelling software</i>	2012-2016
Worked on dynamic modelling, participated in two Google Summer of Code (GSOC) projects as a student (2012, 2013) and in two GSOC projects as a mentor (2015, 2016)	

LANGUAGES

- English, German: fluent
- Russian: mother tongue